



Simulation and experiment of a dual-temperature zone MBE heater

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ABSTRACT

Achieving precise thermal control over large substrates remains a critical challenge in molecular beam epitaxy (MBE) systems, where conventional single-temperature zone heaters often fail to deliver uniformity under high-temperature and ultra-high vacuum conditions. This study introduces a novel dual-temperature zone heater architecture for large-scale MBE, integrating multi-layer insulating and reflecting plates to address non-uniform heat distribution. A coupled electrical–thermal–mechanical simulation framework is established to capture multi-physics interactions, enabling structure optimization beyond traditional empirical tuning. Furthermore, a predictive modeling approach systematically links geometric parameters to thermal uniformity, providing a quantitative basis for refining the design. Experimental results show that the heater maintains a substrate temperature difference within ± 3 K at the target temperature of 983 K, with a maximum achievable temperature of 1275 K and a simulation-to-experiment maximum deviation of 2.4 K. These findings confirm the reliability of the proposed co-design methodology and demonstrate a substantial improvement in uniformity compared with conventional designs. The proposed strategy offers a scalable framework for next-generation high-precision thermal control in MBE and related large-scale deposition systems.

1. Introduction

The rapid growth of high-end equipment manufacturing has increased the demand for precise growth and thermal control of micro/nano structures. Consequently, film growth technologies are becoming essential for precision instrumentation [1,2], aerospace [3–5], and optoelectronic devices [6–9]. As an advanced epitaxial technology, MBE is indispensable for semiconductor fabrication and represents the frontier of film preparation [10]. With advantages such as low-temperature growth [11,12], high cleanliness, and precise control, MBE is well-suited for the fabrication of ultrathin quantum structures and superlattices [13]. It enables the fabrication of advanced structures such as two-dimensional electron gas (2DEG) [14], multiple quantum wells [15], and quantum wires or dots [16–19]. MBE is widely used in fabricating terahertz devices [20], quantum cascade lasers, VCSELs [21], optical waveguides, and other high-performance optoelectronic devices [22]. It also plays a key role in high-end manufacturing equipment [23, 24]. Emerging industries such as 5 G, artificial intelligence, and IoT demand higher quality, uniformity, and thermal control of MBE-grown semiconductors. These demands drive continuous improvements in

high-precision thermal control systems [25,26].

Temperature control in MBE, particularly heater performance, has become a significant research focus at the intersection of mechanical engineering and precision thermal control [27,28]. In high-precision thermal control and material fabrication, temperature accuracy directly determines the quality of functional structures and device performance [29–32]. For MBE systems, substrate temperature critically affects atomic migration [33], interfacial reactions [34], nucleation, micro structure [35], defect evolution [36], and material properties [37]. Arimoto et al. [38] reported its significant effect on surface morphology and defect formation, while Cevher et al. [39] observed that higher temperatures increase minority and non-radiative carrier lifetimes. Other studies confirm its strong influence on surface roughness [40,41], thickness [42–45], and alloy composition [46–51]. Marconi et al. [52] achieved stepwise growth of Au films on Si(111) at 580 °C. They noted increased surface roughness with temperature, with a minimum roughness of 0.32 nm under monolayer-by-monolayer growth conditions. Liu et al. [53] reported optimal 3C-SiC film quality at 1000 °C. Excessive temperature increased dislocations due to thermal expansion mismatch, while low temperature caused compositional

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deviation and impurity accumulation, reducing crystal integrity and device performance. However, most studies focus on single-parameter adjustments and lack a unified thermo-mechanical optimization framework, which limits temperature uniformity under complex conditions.

The scaling-up and precision fabrication of semiconductor devices have made MBE heater design a critical bottleneck for thermal stability and film uniformity. Recent advances in gas-phase epitaxy provide insights into heat source design. Tavakoli et al. [54] analyzed coil geometry and spatial arrangement, proposing a comparative design framework. Huang et al. [55] introduced a multi-layer coil to reduce temperature deviation within 5 K. Qu et al. [56] examined heating wire distribution and recommended a “loose-inside, tight-outside” strategy to mitigate edge overheating. These studies [57–59] established foundational simulation methods for optimizing heat sources. However, MBE systems under ultra-high vacuum and radiation-dominant conditions pose greater challenges. Researchers have addressed these challenges by developing specialized radiant heaters. Finegan et al. [60] achieved ± 8 K uniformity at 1073 K for 2-inch substrates. Shengurov et al. [61] achieved operation up to 1723 K on sapphire substrates, while Rogers et al. [62] reduced nonuniformity from >20 K to 6.5 K via reflecting design adjustments.

Despite progress in small-scale MBE heater optimization, most studies treat thermal control and structural design separately, lacking an integrated framework for large-scale, multi-zone architectures. Large-area MBE heaters must ensure temperature accuracy, spatial uniformity, and mechanical robustness, requiring a shift from single-parameter tuning to integrated multiphysics co-design [63–65]. Thermal control is evolving from isolated heat source regulation to holistic multi-source synergy, considering radiation boundaries, reflecting geometries, and structural layouts as coupled variables [66–68]. However, the complex thermal response of large substrates remains insufficiently modeled and optimized, posing a significant challenge for next-generation MBE systems. Mechanical sciences offer mature tools for coupled thermal-mechanical simulation using FEM, multi-physics platforms (e.g., COMSOL, ANSYS), and experimental validation [69–80]. Applying these methods to MBE heater design allows accurate thermal field prediction under high-temperature and vacuum conditions. It quantifies structural effects, enabling a shift from pure thermal to thermal-mechanical design. This provides a basis for new structures such as integrated heat transfer systems and heaters with low-stress frames [81–83]. The growing complexity of thermal control makes empirical design inadequate to meet the accuracy and uniformity demands of large MBE systems. Multi-parameter optimization methods [84–88] have become crucial in addressing multivariate, nonlinear, and coupled heater design problems, enabling the systematic evaluation of structural parameter combinations for optimal performance. This opens the path to multi-physics co-design of MBE heater thermo-mechanical structures. However, the main barrier to achieving industrial-level uniformity and stability is the lack of an integrated thermo-mechanical modeling and optimization framework for large-scale, multi-temperature zone MBE heaters.

To overcome the limitations of current studies, including restricted parameter ranges and unclear thermo-structural response mechanisms, this study proposes a novel thermal design paradigm for a 4×6 -inch MBE heater. The core innovation lies in a dual-temperature zone heater architecture combined with a multi-physics and multi-parameter co-design framework, which departs from traditional single-parameter tuning by enabling structure-level optimization under ultra-high vacuum conditions. A coupled electrical-thermal-mechanical simulation model captures the complex interactions between the heating plate, reflecting structures, and substrates. An integrated optimization strategy systematically links structural parameters to thermal uniformity. Unlike previous methods, this closed-loop approach integrates simulation, optimization, and experimental validation, resulting in superior uniformity and stability. Beyond addressing MBE design bottlenecks, this

methodology establishes a scalable foundation for next-generation high-precision thermal control systems. The paper is organized as follows: Section 2 details the heater structure and simulation model, Section 3 presents parameter optimization and experimental validation, and Section 4 summarizes key findings.

2. Numerical model

This section begins with a detailed description of the MBE heater structure and heating principles. It then outlines the simulation assumptions and numerical methods and summarizes the energy conservation equations and boundary conditions. Finally, the heater structure optimization process, simulation model construction, and validation are presented.

2.1. Physical model

This subsection introduces the physical model of the MBE thermal management system, focusing on its structure, operating principle, and material properties. These details provide the basis for the simulations and analyses presented in later sections.

The heating system of the 4×6 -inch MBE apparatus comprises a heater and a cold trap. Fig. 1(a) illustrates the 3D layout of the system, emphasizing the spatial arrangement between the heater and the cold trap. Fig. 1(b) and Fig. 1(c) show the heater's cross-sectional view and enlarged details, clearly depicting the assembly of its components. Fig. 1(d) provides a detailed breakdown of all heater components. During operation, electric current flows from the wiring terminal to the pyrolytic graphite (PG) heating plate, generating heat via the Joule effect and rapidly raising the plate's temperature. Due to the thermal gradient between the heating plate and substrate, heat is mainly transferred by thermal radiation, effectively heating the substrate.

As the temperature increases, the thermophysical properties of PG vary significantly due to enhanced atomic and molecular activity, which alters the lattice structure and defect distribution. Additionally, temperature changes affect the behavior of electrons and phonons, impacting graphite's thermal conductivity. Therefore, it is crucial to account for temperature-dependent thermophysical properties of PG in simulations. Specifically, critical parameters, including thermal conductivity, electrical conductivity [89,90], and specific heat capacity [91, 92], vary with temperature, as shown in Fig. 2. The properties of other primary structural materials are summarized in Table C1.

2.2. Mathematical model

This subsection proposes five hypotheses applicable to this simulation based on the process conditions of MBE heaters in the real world. Second, the energy control equations in solid and radiation heat transfer processes were introduced based on the coupling of multiple physical fields.

2.2.1. Assumptions

Before performing the simulation, we applied essential simplifications and established key assumptions to represent real-world conditions while concentrating on the core research objectives.

- (1) Neglect of convective heat transfer: Given that the MBE system operates under ultra-high vacuum conditions during operation, the effect of convective heat transfer was disregarded. Additionally, radiation absorption by the medium was ignored, considering only thermal radiation and conduction as the dominant heat transfer mechanisms.
- (2) Constant vacuum pressure during substrate heating: Although the molecular beam flux lowers the vacuum level in the chamber during deposition, the substrate is first heated to the target temperature before beam flux introduction. Therefore, the

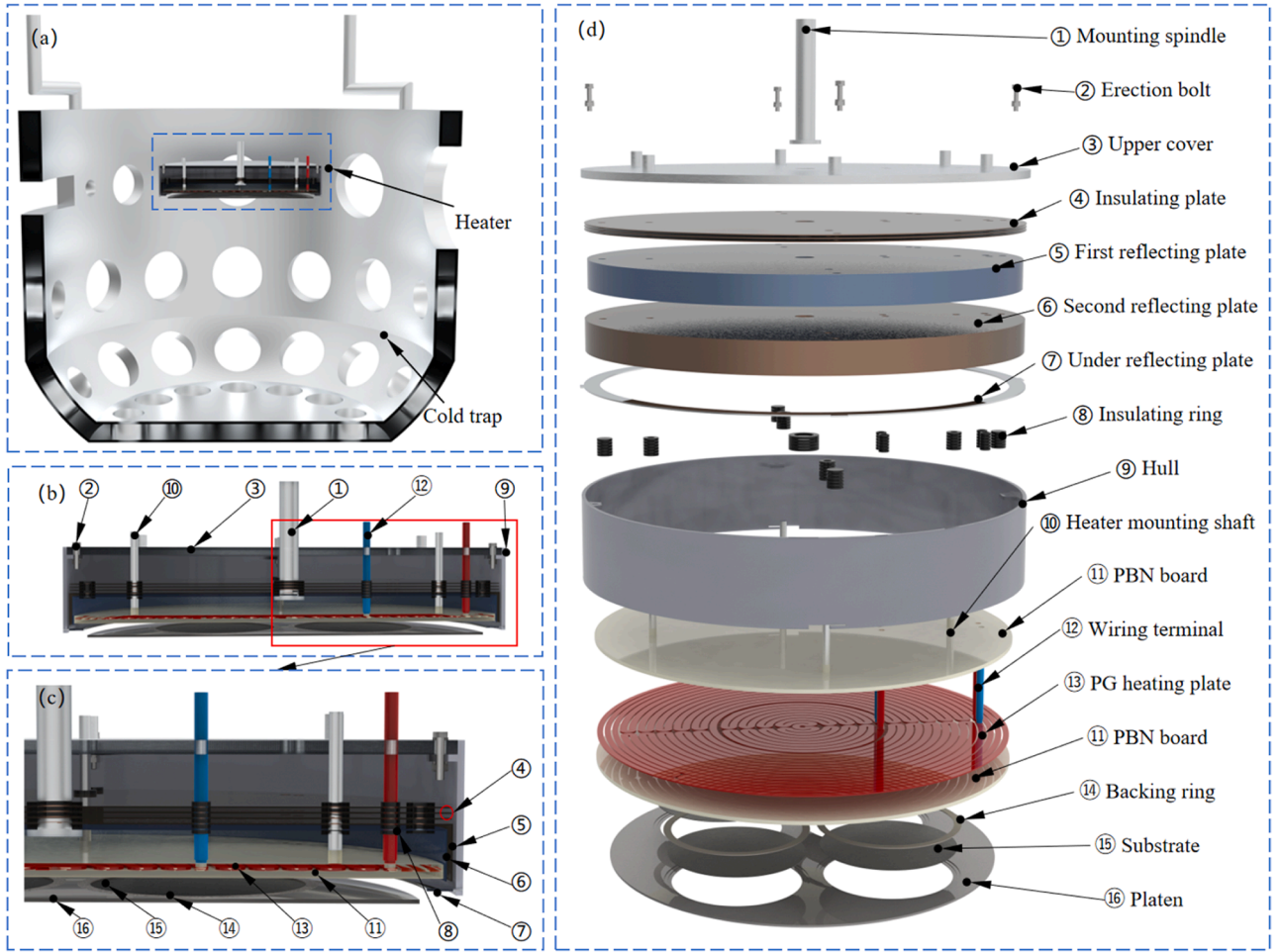


Fig. 1. Physical model of the dual-temperature zone heating system for a large-scale MBE system. (a) Overall 3D layout of the heater assembly, illustrating the spatial arrangement of significant components. (b) A sectional view of the heater model reveals its internal structure, including the configuration of heating elements, insulating layers, and support structures. (c) The enlarged localized view focuses on critical interface regions, providing a detailed visualization of component connections. (d) An exploded view of the heater model, systematically displaying all constituent parts and their assembly relationships.

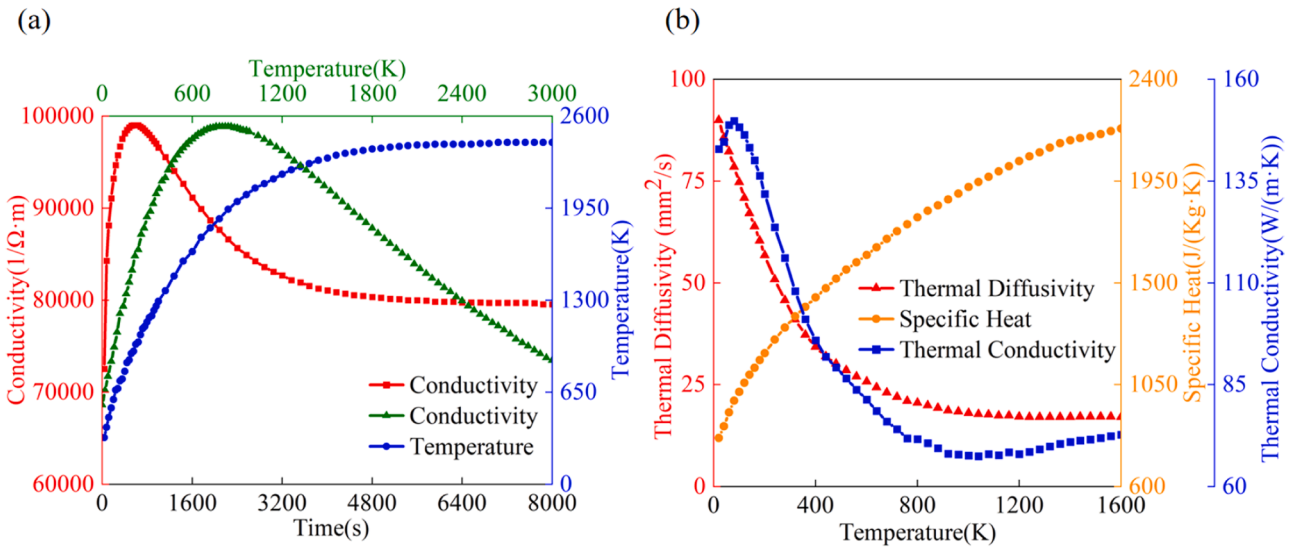


Fig. 2. PG's material properties. (a) The red and olive green curves represent the changes in PG conductivity over time and temperature, respectively, while the blue curve describes the changes in PG temperature over time [89,90]. (b) Thermophysical properties of PG [91,92]. The red curve illustrates the variation of thermal diffusivity with temperature, the orange curve shows the increase in specific heat capacity as the temperature rises, and the blue curve represents the change in thermal conductivity with temperature.

chamber pressure was assumed to remain constant during substrate heating, disregarding the effect of pressure fluctuations.

- (3) Heater material assumptions: The heater materials were modeled as opaque solids and diffuse gray bodies, assuming no thermal radiation penetration through material surfaces. The surface absorptivity and emissivity were assumed equal, and the surfaces were treated as diffuse reflectors, excluding specular reflection.
- (4) Substrate deformation and rotation: The model did not account for substrate deformation at different temperatures or include substrate rotation during heating.
- (5) Steady-state temperature distribution: Only the substrate's steady-state temperature distribution was analyzed.

2.2.2. Governing equations

During the simulation, a multi-physics coupling model integrating Electric Currents, Heat Transfer in Solids, and Surface-to-Surface Radiation was developed using COMSOL Multiphysics 6.1. When current flows through the heating plate, it rapidly generates heat via the Joule effect. Under ultra-high vacuum conditions, heat was primarily transferred to the substrate via thermal conduction and radiation, facilitating substrate heating. During this process, the energy conservation equation for solids is expressed as follows [93]:

$$\rho C_p \partial T / \partial t + (q_{con} + q_{rad}) = Q_{joule}, \quad (1)$$

$$q_{con} = -\lambda \nabla T, \quad (2)$$

$$Q_{joule} = J \cdot E = \sigma \cdot |\nabla V|^2. \quad (3)$$

where ρ is the solid density; C_p is the heat capacity; T is the temperature; t is the time; q_{con} is the conducted heat flux; q_{rad} is the radiative heat flux; Q_{joule} is the Joule heat; λ is the thermal conductivity; E is the electric field strength; V is the electric potential; J is the current density and σ is the conductivity.

In the MBE system, the radiative heat flux (q_{rad}) can be calculated using Eq. (4).

$$q_{rad} = \varepsilon (G - n^2 \sigma_{SB} T^4), \quad (4)$$

$$G = G_m + F_{amb} e_b \varepsilon_{amb} (T_{amb}) + \sum F_{ext} P_s + \sum F_{ext} q_{0,s}. \quad (5)$$

where G is the incident flux, e_b is the radiosity; n is the refractive index, G_m is the mutual radiation between different boundaries, F_{amb} is the ambient view factor, F_{ext} is the external view factor, P_s is the aggregate sum of radiation sources located at a point, $q_{0,s}$ are directional radiation sources, ε is the surface emissivity, ε_{amb} is the ambient emissivity; T is the temperature; T_{amb} is the ambient temperature; σ_{SB} is Boltzmann's constant.

This model utilizes the Hemi-cube algorithm to calculate surface-to-surface radiation. As shown in Fig. 3, a point P_1 exists on surface A_1 , while another surface A_2 and its surroundings allow for observing point P_2 from any location. The view factor of radiation from the A_1 face to the A_2 face can be calculated by the following equation:

$$F_{1,2} = 1 / A_1 \int_{A_1} \int_{A_2} \cos \theta_1 \cos \theta_2 / \pi r^2 dA_1 dA_2, \quad (6)$$

Assuming that the energy E_1 of A_1 is higher than the energy E_2 of A_2 , the net radiation between A_1 and A_2 can be calculated by the following equation:

$$G_m = (E_1 - E_2) \int_{A_1} \int_{A_2} \cos \theta_1 \cos \theta_2 / \pi r^2 dA_1 dA_2, \quad (7)$$

The ambient view factor, F_{amb} , can be calculated by Eq. (8):

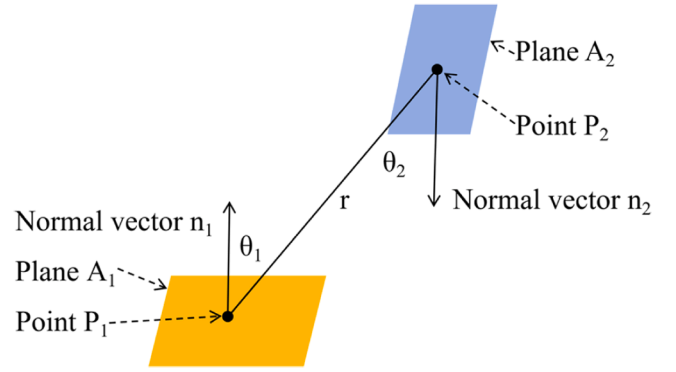


Fig. 3. Schematic of radiative heat exchange between two arbitrary differential surface elements. A_1 and A_2 represent two planes, with P_1 and P_2 denoting points on each plane, respectively. The vector r connects P_1 and P_2 , while n_1 and n_2 are the normal vectors at these points. θ_1 and θ_2 are the angles between the normal vectors (n_1 , n_2) and the line segment P_1P_2 .

$$F_{amb} = 1 - F_{1,2} = 1 - 1 / A_1 \int_{A_1} \int_{A_2} \cos \theta_1 \cos \theta_2 / \pi r^2 dA_1 dA_2. \quad (8)$$

2.3. Optimization process

This subsection introduces the key structural parameters of the substrate heater and presents a simulation-based optimization process. The optimization is carried out in stages to refine the heater structure and improve substrate temperature uniformity.

Scholars have conducted in-depth studies on heater structures from various perspectives, including heating coil distribution [60], the number of temperature zones [94], heating coil layers [55], heating plate spacing [95,96], heating plate shape [95,97,98], insulating structure [99], heating distance [96,100], and platen design [101,102]. Studies have demonstrated that heater structural design is crucial in substrate temperature uniformity.

Similarly, in MBE systems, key parameters, including heating plate configuration, reflecting and insulating plate design, heating zone width and count, and heating spacing, are essential in determining heater performance and substrate temperature distribution. However, studies on substrate heating in large-scale 4×6 -inch MBE systems are relatively limited. Therefore, further research is urgently required to systematically investigate heater design and critical parameter effects on substrate temperature uniformity, facilitating the optimization of heating systems in large-scale MBE equipment.

The key parameters of the designed heater are illustrated in Fig. B1 in Appendix B. Specific as follows: heating plate parameters: width of heating plate (d), spacing of heating plate (l), and thickness of heating plate (w); insulating plate parameters: number of insulating plates (N_1), spacing of insulating plate (h_1); reflecting plate parameters: number of reflecting plates (N_2), spacing of reflecting plates (h_2), and width of under reflecting plate (L); temperature zone parameters: width of inner temperature zone (L_A) and width of outer temperature zone (L_B); heating distance (H). The values of these parameters are summarized in Table C2 in Appendix C.

Fig. B2 in Appendix B presents the optimization process for the substrate heater. By adjusting key parameters, including the heating plate structure, insulating/reflecting configurations, and the number and width of temperature zones and heating spacing, this study systematically examines their impact on maximum substrate temperature and temperature uniformity.

The optimization process consists of two stages. First, a single-temperature zone heater model is developed. The optimization sequence starts with the heating plate structure, followed by the insulating and reflecting structures, evaluating their effect on substrate

heating. Each subsequent parameter optimization uses the structure obtained from the previous step as the initial configuration, ensuring a stepwise refinement of the design. After optimizing the single-temperature zone model, a dual-temperature zone heater model consisting of inner and outer temperature zones was developed. The widths of the dual-temperature zone, the structure of the width of the under reflecting plate, and the heating spacing are optimized in sequence. Ultimately, the structural optimization for the 4×6 -inch MBE heater was completed.

Subsequently, the optimized design is compared with the initial model to evaluate improvements. A 4×6 -inch MBE heating system is constructed and experimentally validated based on the refined heater parameters. A comparative analysis of the experimental and simulation results is performed to validate the accuracy of the simulation model.

2.4. Simulation model

This subsection outlines the heat transfer mechanisms and meshing strategy in COMSOL. Under ultra-high vacuum, only conduction and radiation are considered. A balanced meshing scheme ensures both accuracy and efficiency.

Under ultra-high vacuum conditions (10^{-9} Pa), where gas molecules are highly sparse, convection is negligible and can be disregarded. Therefore, only heat conduction and radiation are considered in the simulation. Fig. 4 illustrates the heat transfer mechanisms during system heating.

Heat is generated through the Joule effect when the heating structure is powered. Reflecting plates minimize heat loss, while insulating plates protect the heater casing and chamber components from excessive heating. The substrate is firmly mounted on the platen using a backing ring. Heat from the high-temperature heating plate is primarily transferred to the substrate via radiation, while conduction and radiation propagate heat within the substrate, platen, and backing ring. Differences in material properties and absorption capacities among these components create temperature gradients at their interfaces, enhancing interfacial heat conduction and accelerating heat transfer from hot to cold regions. The heater casing further improves thermal efficiency by reducing heat dissipation, and the under reflecting plate mitigates edge effects that compromise substrate temperature uniformity. Surrounding the system, the cold trap lowers the chamber wall temperature. It adsorbs residual gas molecules to maintain an ultra-high vacuum, thereby reducing molecular collisions on the substrate surface and improving film quality. At the same time, it limits heat transfer from the heater to the chamber walls, optimizing overall thermal management.

Fig. B3 in Appendix B shows the meshing strategy for the simulation

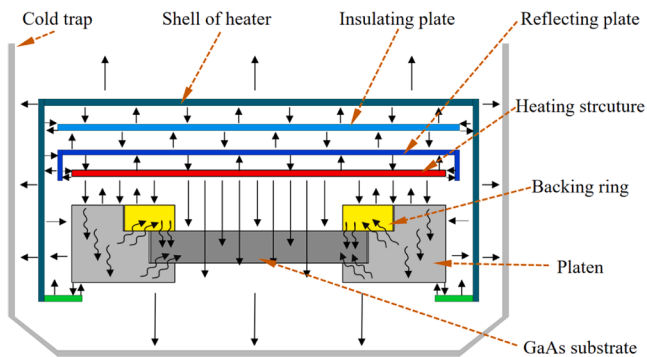


Fig. 4. Schematic diagram of substrate mounting and heating in the MBE system. The diagram illustrates the primary heat transfer pathways: straight black arrows indicate radiative heat exchange between the heater surfaces, substrate, and surrounding reflecting/insulating components, while wavy black arrows denote conductive heat flow within solid components (such as the platen, backing ring, and substrate) and across their contact interfaces.

model in COMSOL Multiphysics 6.1. Hexahedral and prismatic meshes are used for cylindrical regions, while tetrahedral meshes are applied to irregular geometries. The mesh comprises approximately 355,000 elements, including 990 prismatic, 27,700 hexahedral, and 326,310 tetrahedral elements. The average mesh quality is 0.692, with a minimum element quality of 0.018. Since the average quality exceeds 0.5, the mesh satisfies computational requirements, ensuring accuracy and efficiency.

2.5. Grid and time-step independence validation

This subsection presents the simulation model's validation of grid and time-step settings. The aim is to ensure computational accuracy and efficiency by identifying suitable grid resolution and time-step size before conducting the main calculations.

When conducting simulations, selecting the appropriate grid division and time-step settings is crucial for balancing computational accuracy and resource utilization. Consequently, validations for grid [103] and time-step independence [104] were conducted before model calculations.

Fig. 5(a) shows the grid independence validation results, covering grid sizes of 50,000, 100,000, 200,000, 350,000, 500,000, and 650,000. Both the maximum temperature and temperature difference stabilize when the grid exceeds 350,000 elements. Therefore, 350,000 grid divisions were selected to optimize computational efficiency and accuracy. Fig. 5(b) presents the time-step independence results, where time steps of 1 s, 6 s, 12 s, 18 s, 24 s, 30 s, and 60 s were analyzed. While the time step size minimizes the maximum substrate temperature, values above 12 s cause significant fluctuations in the substrate temperature difference, reducing accuracy. A time step of 6 s was adopted to ensure accuracy while minimizing computational cost.

3. Results and discussion

This section begins by optimizing the single-temperature zone model using an orthogonal simulation method to evaluate the influence of heating plate structure and insulating/reflecting plate configurations on substrate temperature distribution. Next, the impact of the dual-temperature zone design, the width of under reflecting plate, and the heating spacing on temperature uniformity is analyzed through simulations. Finally, experimental and simulation results at different heating temperatures are compared to validate the multi-physics coupling model.

3.1. Orthogonal design of heating plate

This subsection examines the impact of heating plate structural parameters on substrate temperature through orthogonal simulations. The aim is to identify how parameters affect temperature uniformity and effective radiating area. Providing a systematic basis for selecting an optimal heating plate configuration for dual-temperature zone MBE systems.

The heating plate has a significant impact on the substrate temperature [95,97,98]. Utilizing orthogonal simulations significantly reduces the number of trials while preserving the analysis of multi-parameter effects on evaluation metrics. An orthogonal experimental design was implemented to investigate the effects of the width of heating plate (d) and the spacing of heating plate (l) on substrate temperature difference and effective radiating area. The factor levels corresponding to these parameters are presented in Table C3 in Appendix C.

A range analysis was conducted on the substrate temperature difference (β) and effective radiating area (γ) under different heating plate parameters. The results are shown in Table C4 in Appendix C, where I_i/J_i ($i = 1, 2, \dots, 5$) represents the sum of all evaluation indices for the i th level of a given factor, with i denoting the number of factor levels. The larger the R -value, the more significant the influence of the

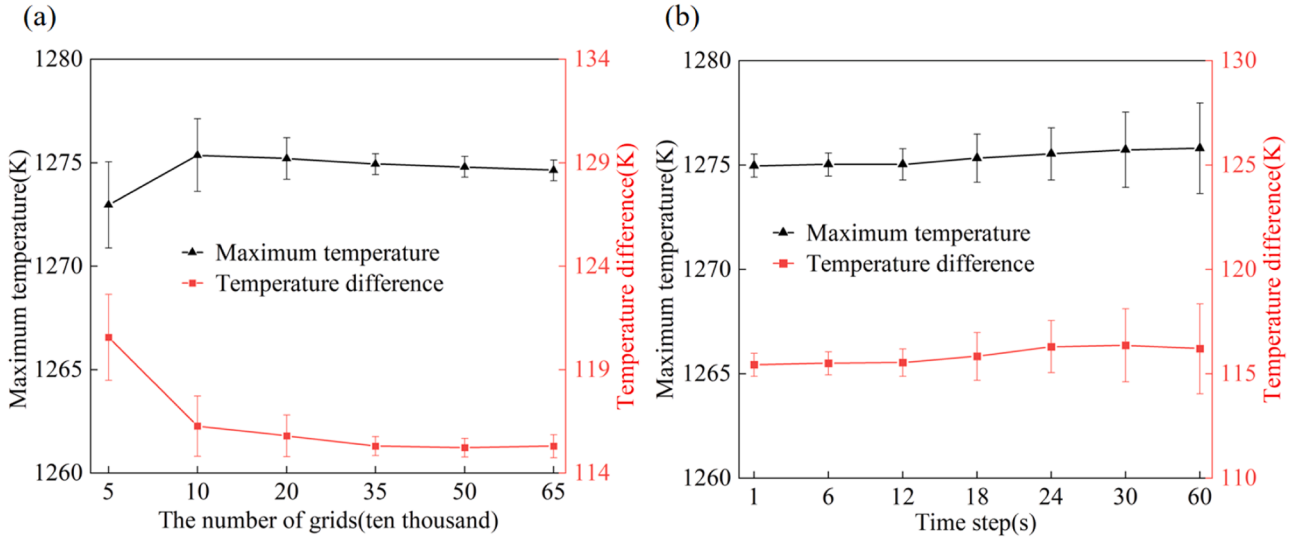


Fig. 5. Validation of numerical model for grid and time-step independence. The black curve represents the maximum substrate temperature, and the red curve represents the substrate temperature difference. (a) Grid independence validation results; (b) Time-step independence validation results.

corresponding parameter on the evaluation metrics. The range results show that for β and γ , $R_f(l) > R_f(d)$ and $R_f(l) > R_f(d)$, l produces a greater effect than d .

Evaluation criteria: when $P < 0.01$, it represents an extremely significant influence, expressed as ***. When $0.01 < P < 0.05$, it represents significant influence, expressed as **. when $0.05 < P < 0.1$, it represents a certain influence, expressed as *. when $P > 0.1$, represents no significant influence, expressed as -. The smaller the P -value, the more pronounced the influence of the test parameters on the test results.

As shown in Table C5 in Appendix C, for β and γ , the P value < 0.01 , and the significance levels follow the order $l > d$. This indicates that the spacing/width of the heating plate significantly impacts the substrate temperature difference and effective radiation area. Notably, the influence of spacing exceeds that of width, highlighting its dominant role in thermal regulation. This trend can be explained physically: the reduced spacing results in a more concentrated heat in the space, with less heat loss, creating a typical multi-body radiation enhancement effect. This increases the thermal efficiency and enhances the radiative coupling

between the heating plates. At the same time, it reduces the gap between neighboring heating plates, thereby increasing the effective radiation area and heat flux density. On the contrary, the change in width mainly affects the effective radiant area and does not significantly improve the energy exchange pattern, resulting in a weak effect. Therefore, spacing plays a more decisive role in governing the spatial distribution of thermal flux and achieving temperature uniformity.

Trends in substrate temperature difference and effective radiating area for different parameters are shown in Fig. 6, based on the data in Table C4 (Appendix C). A multivariate nonlinear regression model describing the relationship between evaluation indices and parameter values was constructed (Appendix D), and predictions based on this model are presented in Fig. 7. For maximizing effective radiating area and enhancing radiative heat transfer efficiency, the configuration with $d = 11$ mm and $l = 1$ mm performed best. The reduced spacing increases view factors and promotes radiation-trapping effects, accelerating heat transfer and improving thermal response. However, in dual-zone heating, where thermal fields of independently controlled regions must

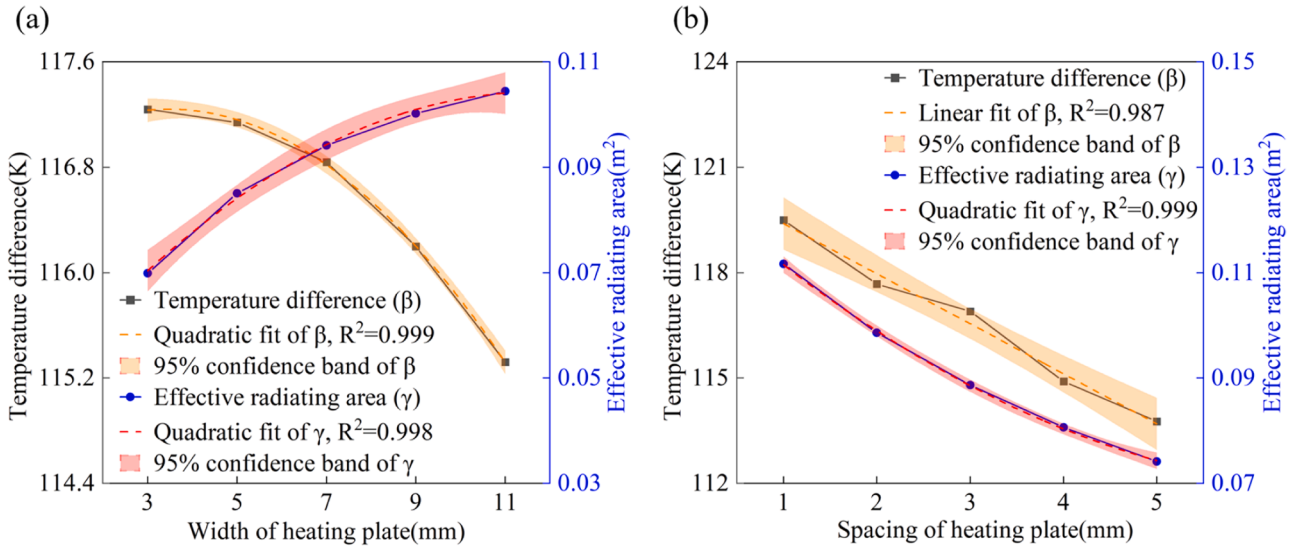


Fig. 6. Influence of heating plate parameters on substrate temperature difference (β) and effective radiating area (γ). In the figure, the dashed line represents the fitting curve, the colored regions of different widths represent the 95 % confidence bands, and R^2 represents the fitting determination coefficient. (a) Describes the trend of β and γ with d . (b) Shows the decreasing trend of β and γ with increasing l .

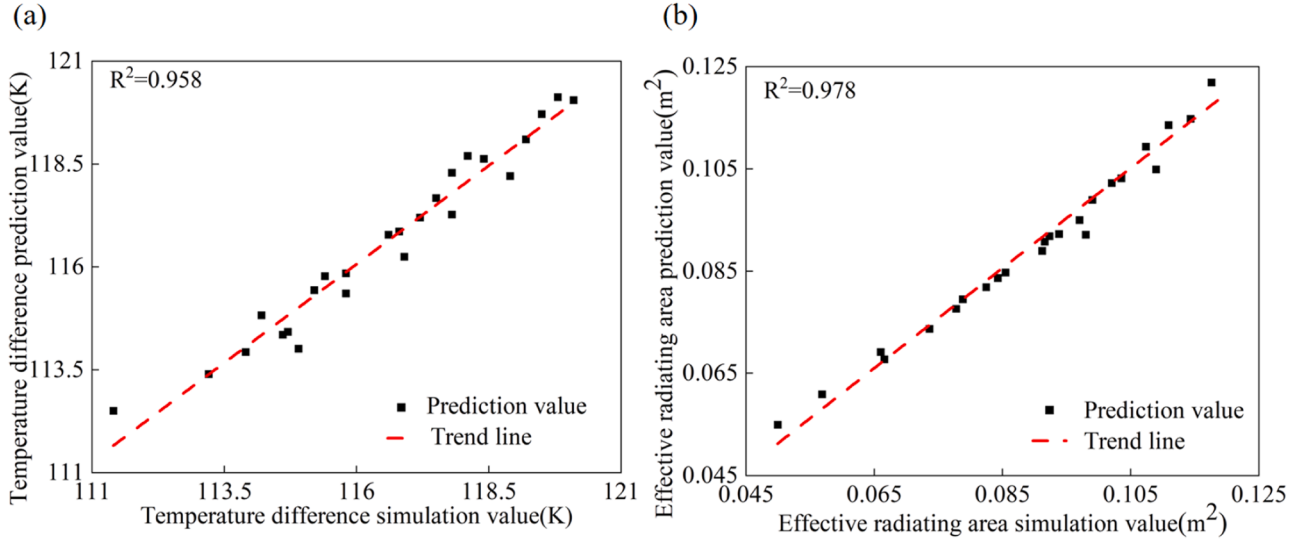


Fig. 7. Comparison of simulation results with regression model predictions. (a) Substrate temperature difference. (b) Effective radiating area. The black scatter points represent the model predicted values. The red dashed line illustrates the fitted trend line of the predicted values relative to the simulated values. The coefficient of determination (R^2) is shown to indicate the goodness of fit.

remain balanced, the configuration $d = 3$ mm and $l = 1$ mm achieved better inter-zone consistency. Narrower plates and smaller spacing enable finer modulation of localized heat input, reducing thermal overspill and ensuring zone-specific setpoints, while minimizing thermal interference.

Considering trade-offs among thermal uniformity, heating efficiency, zone independence, and manufacturability, a final configuration of $d = 9$ mm and $l = 2$ mm was selected. This design minimizes temperature differences, maintains a large radiating area, and ensures robust dual-temperature zone coordination.

3.2. Orthogonal design of insulating plates and reflecting plates

This subsection investigates the effects of insulating and reflecting plates on the MBE system's substrate and heater shell temperature. Orthogonal simulations are used to analyze key parameters through trend plots, F -tests, and physical mechanism interpretation.

Insulating and reflecting plates are critical components of heating systems [95,96,99]. The insulating plate serves two key functions: (1) isolating the heating zone from the chamber environment to maintain temperature stability; (2) protecting chamber components by lowering heater surface temperature, preventing heat-induced damage, and improving thermal safety. The reflecting plate fulfills three primary roles: (1) redirecting thermal radiation to minimize heat loss and improve energy utilization; (2) reducing heat dissipation to maintain heating zone stability and extend heater service life; (3) promoting uniform heat distribution within the heating zone, thereby minimizing temperature gradients and enhancing thermal uniformity.

An orthogonal experiment evaluated the effects of reflecting and insulating plates on the MBE system's substrate temperature difference and average shell temperature. The factor levels are provided in Table C6 (Appendix C). Based on the experimental results, trend plots were generated to show how the number and spacing of reflecting and insulating plates influence the substrate temperature difference (β) and

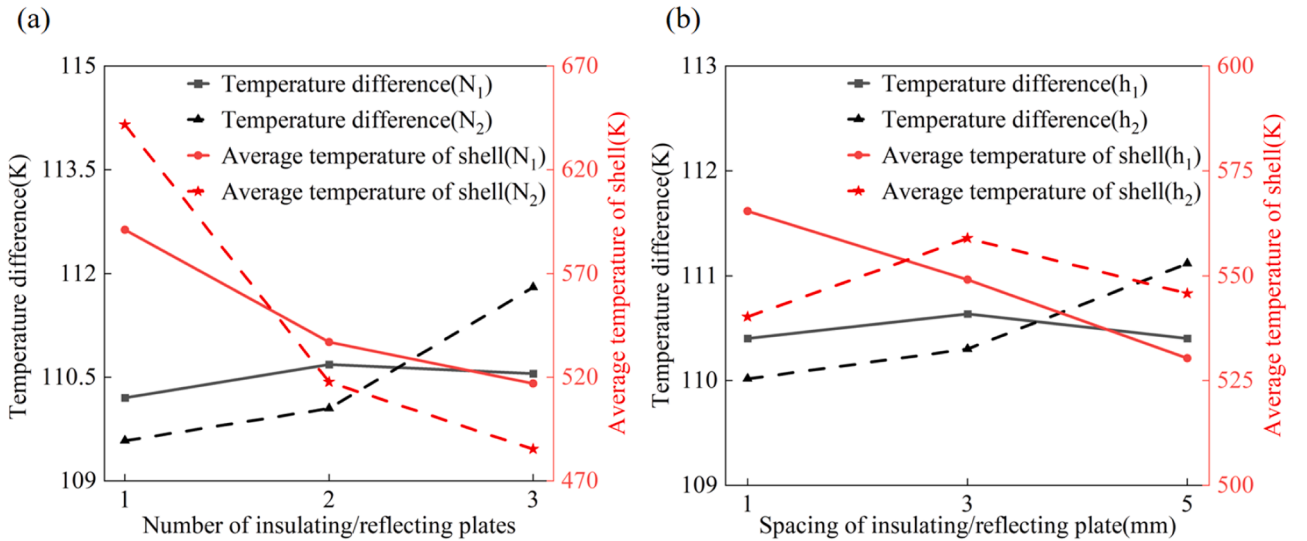


Fig. 8. Influence of key parameters on substrate thermal performance metrics. Where the black line depicts the substrate temperature difference β versus the parameter, and the red line shows the average shell surface temperature δ trend versus the parameter. (a) The effect of N_1/N_2 on β/δ is depicted. (b) Demonstrates the effect of different values of h_1/h_2 on β/δ .

the average temperature of shell (δ), as illustrated in Fig. 8. These effects were further analyzed using an F -test to determine the statistical significance of each parameter.

As shown in Table C7 (Appendix C), the results indicate that the number of reflecting plates (N_2) and the spacing of reflecting plate (h_2) have a highly significant effect on β , with significance levels following the order $N_2 > h_2$. N_2 also strongly influences δ , while the number of insulating plates (N_1) has a notable effect on δ , with overall significance ranked as $N_2 > N_1$. Other parameters show no significant impact on β or δ . In addition, Fig. 8 reveals that changes in N_1 and the spacing of insulating plate (h_1) alter β by <0.5 K, which is negligible compared to the overall temperature difference of 110 K.

Beyond the F -test, a deeper thermophysical interpretation reveals that reflecting and insulating plate configurations jointly influence radiation path confinement, energy redistribution, and structural coupling. Increasing the number of reflecting plates enhances radiant heat return, concentrating heat near the substrate. However, it also complicates radiation paths, causes uneven heat flux distribution, and ultimately increases the substrate temperature difference. In comparison, the spacing of reflecting plates has a negligible effect, as moderate changes in spacing have a minimal influence on radiation angles and heat flux distribution. The insulating plate impacts the heat conduction path: increasing the number of insulating plates raises the system's overall thermal resistance, reduces heat transfer to the shell, and significantly lowers the average temperature of shell, while exerting little influence on substrate heating. Adjusting to reflecting and insulating plates collectively forms a coupled thermal resistance network that governs the distribution of heat and energy loss. Thus, optimal heater design requires a multi-objective trade-off among energy efficiency, thermal uniformity, and structural complexity.

Based on the orthogonal simulation results, optimal parameter combinations were evaluated using β and δ as primary performance metrics. The lowest β was observed for the configuration $N_1 = 1$, $h_1 = 1$ mm, $N_2 = 1$, $h_2 = 1$ mm. However, under this setting, δ reached 737.11 K, indicating significant heat leakage to the shell, which poses long-term thermal stability concerns. In contrast, the configuration $N_1 = 3$, $h_1 = 5$ mm, $N_2 = 3$, $h_2 = 1$ mm achieved the lowest δ , attributed to the extended thermal resistance path formed by thicker and more numerous insulating plates, effectively limiting conductive and radiative heat transfer to the shell. Nonetheless, this configuration induced a substantial rise in β due to increased internal thermal resistance and weakened radiative feedback, disrupting the substrate's heat uniformity.

To balance thermal uniformity and energy confinement while considering the physical constraints of heater assembly, the optimal configuration was determined as $N_1 = 3$, $h_1 = 5$ mm, $N_2 = 2$, $h_2 = 1$ mm. This setup offers a compromise between sufficient thermal insulation and effective radiative recycling. Three insulating plates with expanded spacing ensure robust suppression of heat loss. In comparison, two reflecting plates maintain a stable and concentrated radiative field toward the substrate without introducing significant thermal gradients.

3.3. Heating plate dual-temperature zone design

This subsection examines the influence of dual-temperature zone configuration on substrate temperature uniformity. Multi-physics simulations assess how inner and outer zone width variations affect radial temperature distribution, uniform area ratio, and substrate thermal homogeneity.

The multi-temperature zone heating structure enables flexible surface temperature control by adjusting the heating power distribution across different zones, allowing for precise substrate temperature regulation [94]. A dual-temperature zone heater model was developed based on the optimized single-temperature zone model.

To analyze the influence of temperature zone width on heating uniformity, the width of inner temperature zones (L_A) and width of outer temperature zones (L_B) were set as follows: ($L_A = 185$ mm, $L_B = 9$ mm),

($L_A = 174$ mm, $L_B = 20$ mm), ($L_A = 163$ mm, $L_B = 31$ mm), ($L_A = 152$ mm, $L_B = 42$ mm), and ($L_A = 141$ mm, $L_B = 53$ mm). The substrate was heated to a maximum temperature of 1275 K, and the resulting temperature distribution was analyzed to assess uniformity.

Multi-physics simulations were conducted using a dual-temperature zone model to evaluate the influence of temperature zone width on substrate temperature uniformity. Among the four substrates, their temperature distributions were nearly identical, and one was selected as a representative sample for detailed analysis. A ± 2 K range was defined as the uniform temperature area, and its spatial distribution under different zone width combinations was visualized (Fig. 9). ImageJ software was employed to quantify the ratio of the uniform temperature area to the effective substrate area, herein referred to as the uniform area ratio. This metric provides a quantitative basis for evaluating the spatial temperature uniformity across the substrate surface.

Further, the radial temperature profiles of the substrate under varying L_A were extracted and visualized (Fig. 11(a)). The corresponding substrate temperature difference and uniform area ratio were plotted (Fig. 11(b)) to explore their relationships with L_A . Compared with the single-temperature zone configuration, the dual-temperature zone heating effectively offsets edge cooling effects through radial compensation. This synergistic control strategy decreases the maximum substrate temperature difference from 117.5 K to below 25 K. The temperature uniformity index increases by approximately 78.7 %, significantly enhancing the overall temperature uniformity of the substrate. As shown in Fig. 10, the temperature uniformity index improved by 93.8 % when $L_A = 174$ mm and $L_B = 20$ mm, respectively.

Notably, the uniformity metrics exhibit a non-monotonic trend with increasing L_A . As L_A increases from a small value, the inner zone gradually dominates central heating, while the outer temperature zone ensures edge temperature stability. This coordination minimizes radial heat flux divergence, leading to improved temperature homogeneity. However, excessive L_A expansion reduces the outer temperature zone's effective compensatory range, leading to thermal imbalance at the periphery. Consequently, the temperature difference and the uniform area ratio exhibit a non-monotonic response, characterized by initial improvement followed by performance degradation beyond the optimal parameter range.

These findings underscore the critical role of the width of the temperature zone configuration in achieving substrate thermal uniformity. The optimal conditions of $L_A = 174$ mm and $L_B = 20$ mm enable precise radial temperature control across the substrate. The system achieves minimal temperature gradients and maximum uniform coverage at this point, reflecting a balanced thermal design. This optimal configuration provides a robust design reference for dual-temperature zone heater development in large-scale MBE systems, where uniform substrate heating is crucial for suppressing thermal stress, ensuring layer thickness homogeneity, and enhancing epitaxy quality.

3.4. The under reflecting plate and heating spacing optimize

In this subsection, the effects of the width of the under reflecting plate (L) and heating spacing (H) on substrate temperature uniformity were investigated. Building on the previously optimized temperature zone widths, a series of simulations and regression analyses was conducted to identify the optimal L and H values.

To further improve substrate temperature uniformity, two key structural parameters, L and H [96,100], were systematically investigated based on the previously optimized dimensions of $L_A = 174$ mm and $L_B = 20$ mm, with the width of under reflecting plate and the heating spacing optimized sequentially. A dual-temperature zone model was established, with L varying from 16 mm to 28 mm in increments of 2 mm and H varying from 10 mm to 20 mm in increments of 1 mm, resulting in a comprehensive set of simulations.

As shown in Fig. 12, both parameters exhibit a distinct parabolic influence on the substrate temperature difference (β). Specifically,

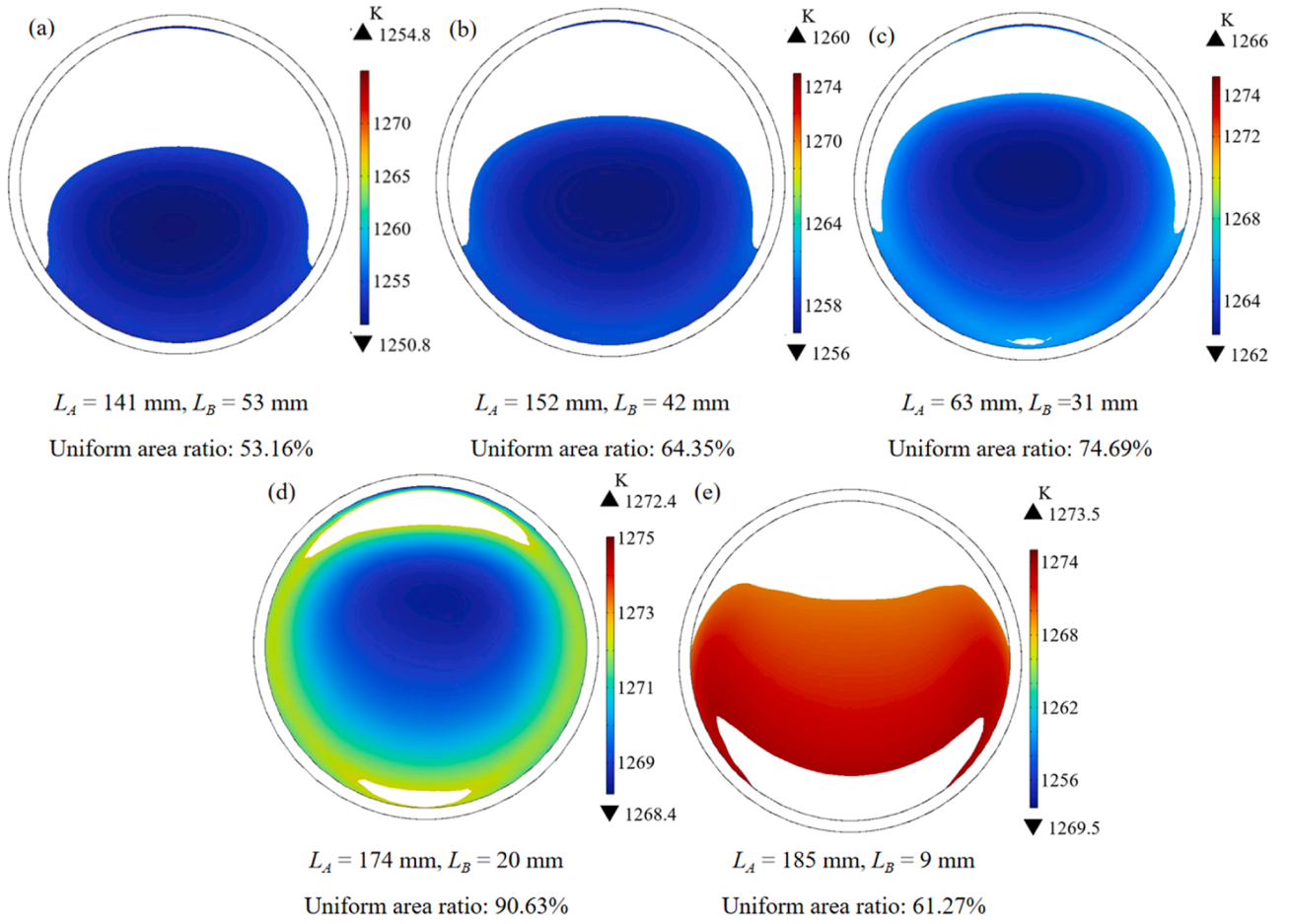


Fig. 9. Effect of temperature zone width configuration on the distribution of the uniform temperature area across the substrate surface. (a-e) The temperature uniform area of the substrate under the conditions of $L_A = 141$ mm, $L_B = 53$ mm / $L_A = 152$ mm, $L_B = 42$ mm / $L_A = 163$ mm, $L_B = 31$ mm / $L_A = 174$ mm, $L_B = 20$ mm / $L_A = 185$ mm, and $L_B = 9$ mm, respectively. The maximum temperature uniform area is obtained when $L_A = 174$ mm and $L_B = 20$ mm.

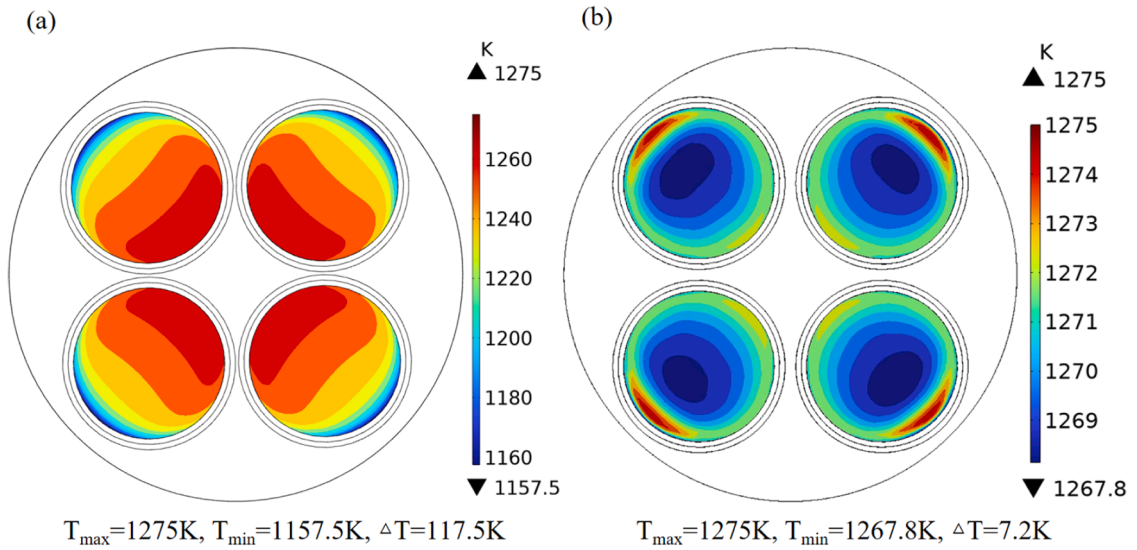


Fig. 10. Comparison of substrate temperature at a target temperature of 1275 K. (a) Single-temperature zone heating without reflecting and insulating plates, showing significant radial non-uniformity. (b) Dual-temperature zone heating with $L_A = 174$ mm and $L_B = 20$ mm, combined with two reflecting plates and three insulating plates. The optimized design markedly improves temperature uniformity, reducing the maximum deviation from 117.5 K in (a) to within 7.2 K in (b).

Fig. 12(a) demonstrates that β initially decreases with increasing L , reaching a minimum at $L = 19$ – 20 mm, after which it rises again. Similarly, Fig. 12(b) shows that β achieves its lowest value when H lies in

the 15–16 mm range. The fitted curves provided in Eqs. (D8) and (D9) in Appendix D demonstrate excellent agreement with the simulation results ($R^2 > 0.98$), with deviations remaining below 5 %, thereby

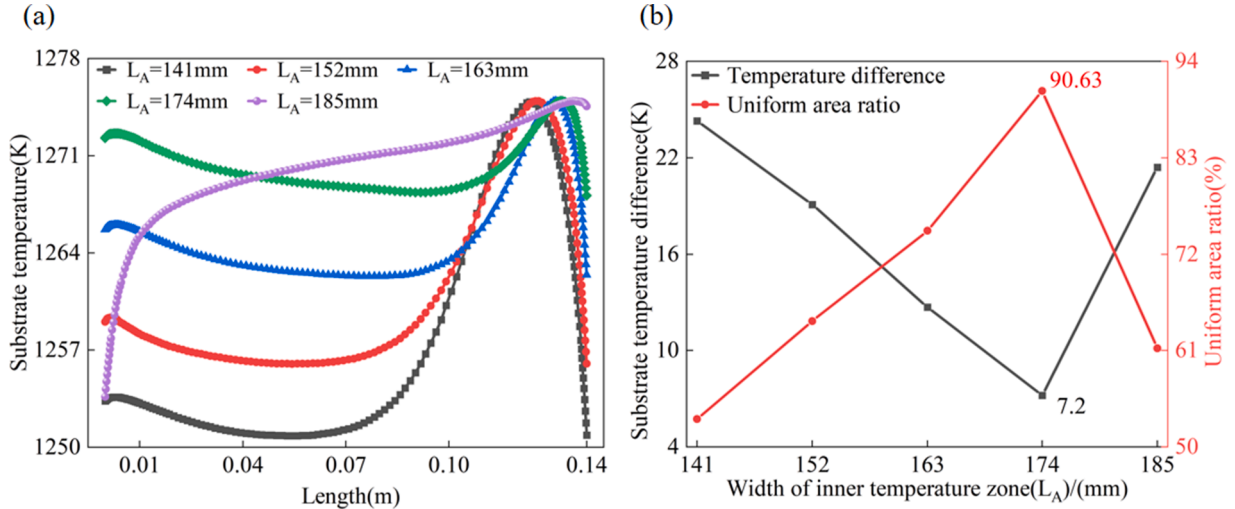


Fig. 11. Effect of the width of inner temperature zones on substrate temperature distribution and uniformity. (a) Radial temperature profiles for five different inner zone widths. (b) Variation of substrate temperature difference (black line) and uniform area ratio (red line) as functions of inner zone width.

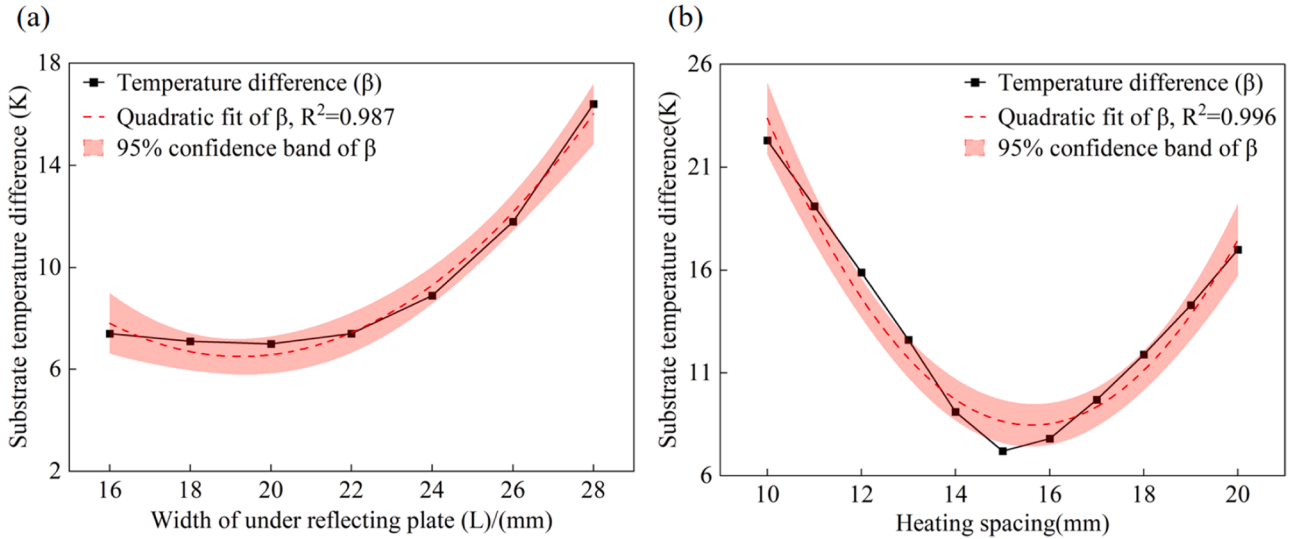


Fig. 12. Effect of heating spacing and width of under reflecting plate on substrate temperature difference. In the figure, the dashed line represents the fitting curve, the colored regions of different widths represent the 95 % confidence bands, and R^2 represents the fitting determination coefficient. (a) Effect of the width of under reflecting plate. (b) Influence of the heating spacing.

confirming the reliability of the numerical model. Additionally, the regression models were subjected to significance-level testing, and P -values were used to determine the statistical significance of the variables' effects on the response. The results are presented in Table C8 (Appendix C). The residual analysis shown in Fig. D1 (Appendix D) confirms that the models exhibit good fitting performance, with reasonably distributed residuals that satisfy the fundamental regression analysis assumptions. Most standardized residuals fall within the range of ± 2 , suggesting minor model errors and an approximately normal distribution. No apparent overfitting behavior was observed, as indicated by the smooth fitted curves and the absence of systematic patterns in the residual plots.

The physical mechanisms underlying these trends can be attributed to competing effects in radiative heat transfer and geometric constraints. For L , increasing the reflecting surface enhances radiative energy confinement and promotes temperature uniformity; however, beyond an optimal threshold, excessive width induces lateral re-radiation losses, reducing the effective radiation flux directed toward the substrate and counteracting thermal homogenization. For H , small gaps ($H < 15$ mm)

increase radiative flux density but concentrate heat in the central region, leading to larger edge temperature gradients due to lateral dissipation. Conversely, considerable spacing reduces overall energy absorption because of geometric divergence, reintroducing non-uniformities.

Therefore, the optimal configuration $L = 20$ mm and $H = 15$ mm effectively balances these competing effects, minimizing thermal gradients and enhancing substrate temperature uniformity. This optimization enhances local thermal homogenization, a critical factor for achieving high-quality epitaxial growth.

3.5. Experimental verification

This subsection presents the experimental validation of the optimized 4×6 -inch MBE heating system. It details the methodology for measuring substrate temperature. Comparisons between simulation and experimental results are provided across target temperatures to verify the simulation model's accuracy and the heater design's effectiveness in achieving high substrate temperature uniformity.

As illustrated in Fig. B4 (Appendix B), the 4×6 -inch MBE system has

a bottom observation window that enables real-time monitoring of its internal operating conditions. During the temperature measurement process, a support frame was positioned below the observation window to secure a Canon EOS 6D Mark II camera, which continuously captured images of the sample stage throughout the entire substrate heating process (Fig. 13) for subsequent image processing and temperature analysis. To enhance the contrast of the substrate region and suppress background light interference, the camera was mounted with a narrow-band red filter with a central wavelength of 660 nm. Additionally, an ND16 neutral density filter was installed to attenuate the incident light intensity without altering the spectral distribution, thereby preventing image overexposure. Infrared image processing was performed using 14-bit RAW grayscale data (0–16,383). Subsequently, multiple calibration experiments were conducted using a FLUKE Endurance infrared thermometer (accuracy: ± 0.1 K) to measure the substrate center temperature and extract the corresponding grayscale values from the substrate images. A numerical mapping relationship between grayscale and temperature was established. Based on this grayscale temperature model, the relative substrate temperature at different heating stages was calculated using image processing results. To verify the reliability of this method, repeated experiments were conducted by comparing the image-based temperature estimates at the substrate center with readings from the FLUKE Endurance infrared thermometer. The results demonstrate that the temperature measurement error remained within ± 2 K, which meets the temperature measurement accuracy requirements.

The designed 4×6 -inch MBE heating system must meet the following performance criteria: once the substrate's maximum temperature reaches 983 K and stabilizes, the substrate temperature variation should be maintained within ± 3 K, while the system must be capable of achieving a maximum heating temperature of 1275 K. To validate the accuracy of the simulation models employed during the heater optimization process and to assess the effectiveness of the heater design, experiments were conducted by heating the substrate to 863 K, 923 K, 983 K, 1043 K, 1103 K, and 1163 K, respectively, while comparing the radial temperature distribution of the substrate. Furthermore, the substrate surface temperature distribution was compared at 983 K to evaluate temperature uniformity and model accuracy.

As illustrated in Fig. B5 in Appendix B, the method for acquiring experimental data on the substrate's radial and surface temperatures is demonstrated. Radial and surface temperature profiles were extracted using a scribing technique on infrared thermal images of the substrate. This method manually drew lines across the thermal images to obtain continuous temperature data for comparative analysis.

As shown in Fig. 14, the simulated and experimental radial temperature distributions exhibit strong consistency in both trend and

magnitude at substrate temperatures of 863 K, 923 K, 983 K, 1043 K, 1103 K, and 1163 K. Fig. 15 presents the comparison of substrate radial and surface temperature differences obtained from simulation and experiment. The maximum deviation between simulated and experimental radial temperature differences is only 0.7 K, while that for surface temperature differences is 2.4 K (Fig. 15(b)). Across the entire temperature range, the relative error between simulation and experiment remains approximately 0.2 %, with all deviations below 5 %, demonstrating the reliability of the numerical model in predicting thermal distribution under various conditions.

At a representative temperature of 983 K, the simulation and experiment exhibit an “M-shaped” radial temperature profile, with the maximum deviation between the two being 0.89 K and a relative difference of < 0.09 %. The experimental and simulated radial temperature differences are 3.1 K and 3.3 K, respectively. Fig. 16 further indicates that the surface temperature distribution follows a saddle-shaped pattern, with the experimental surface temperature difference being 5.3 K compared to 3.3 K from the simulation. The resulting deviation of 2.0 K corresponds to a relative error of < 0.2 %. These findings confirm the system satisfies the ± 3 K temperature uniformity requirement at the target operating conditions.

The high degree of agreement between simulation and experiment underscores the robustness of the developed thermal model. In simulation, the substrate surface and radial temperature differences are identical, and the temperature extrema occur along the radial direction, aligning with theoretical expectations (Fig. 15(a)). In contrast, experimental results reveal slightly larger surface temperature differences than radial differences. This discrepancy primarily originates from the data acquisition method. In the experiments, temperature values were derived from substrate images by manually drawing sampling lines, where the line positions were constrained by the platen structure, resulting in minor deviations in locating the maximum temperature (Fig. 14(a), (b)).

Despite these limitations, the experimental data sample approach employed a dense grid of sampling lines across edge regions to reduce local errors. Nevertheless, manual line extraction inevitably introduces localized uncertainties. By comparison, the simulations utilize a fine computational mesh that provides full-field temperature extraction, ensuring superior spatial resolution and consistency. These methodological differences account for the observed discrepancies while confirming that both approaches converge within acceptable error bounds.

Overall, the experimental results confirm the validity of the simulation model across a wide range of temperatures. The multi-physics simulation framework developed using COMSOL accurately captures the substrate's radial and surface temperature characteristics. This

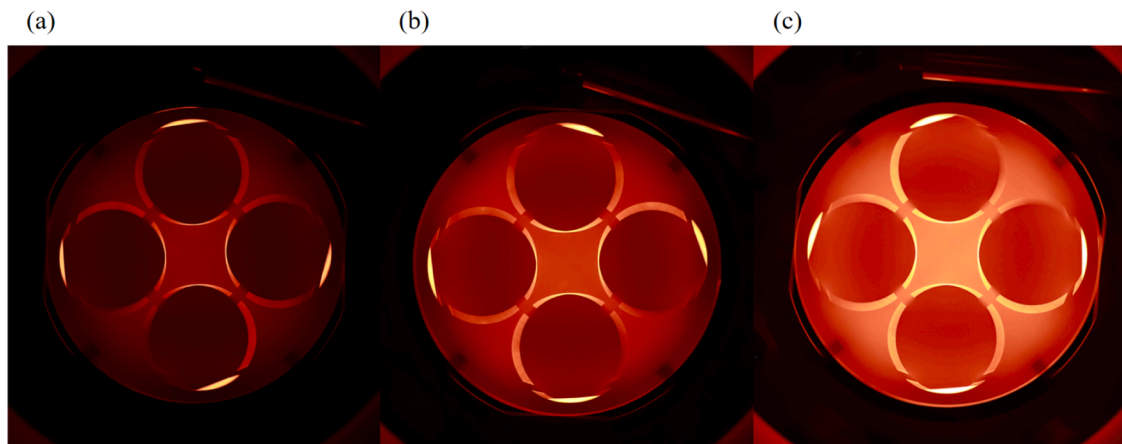


Fig. 13. Infrared thermal images of the substrate under different target temperatures. (a) 983 K, (b) 1043 K, and (c) 1103 K: Each image shows the spatial temperature distribution across the substrate surface under corresponding heating conditions. The infrared images were captured to evaluate surface temperature uniformity and thermal field consistency.

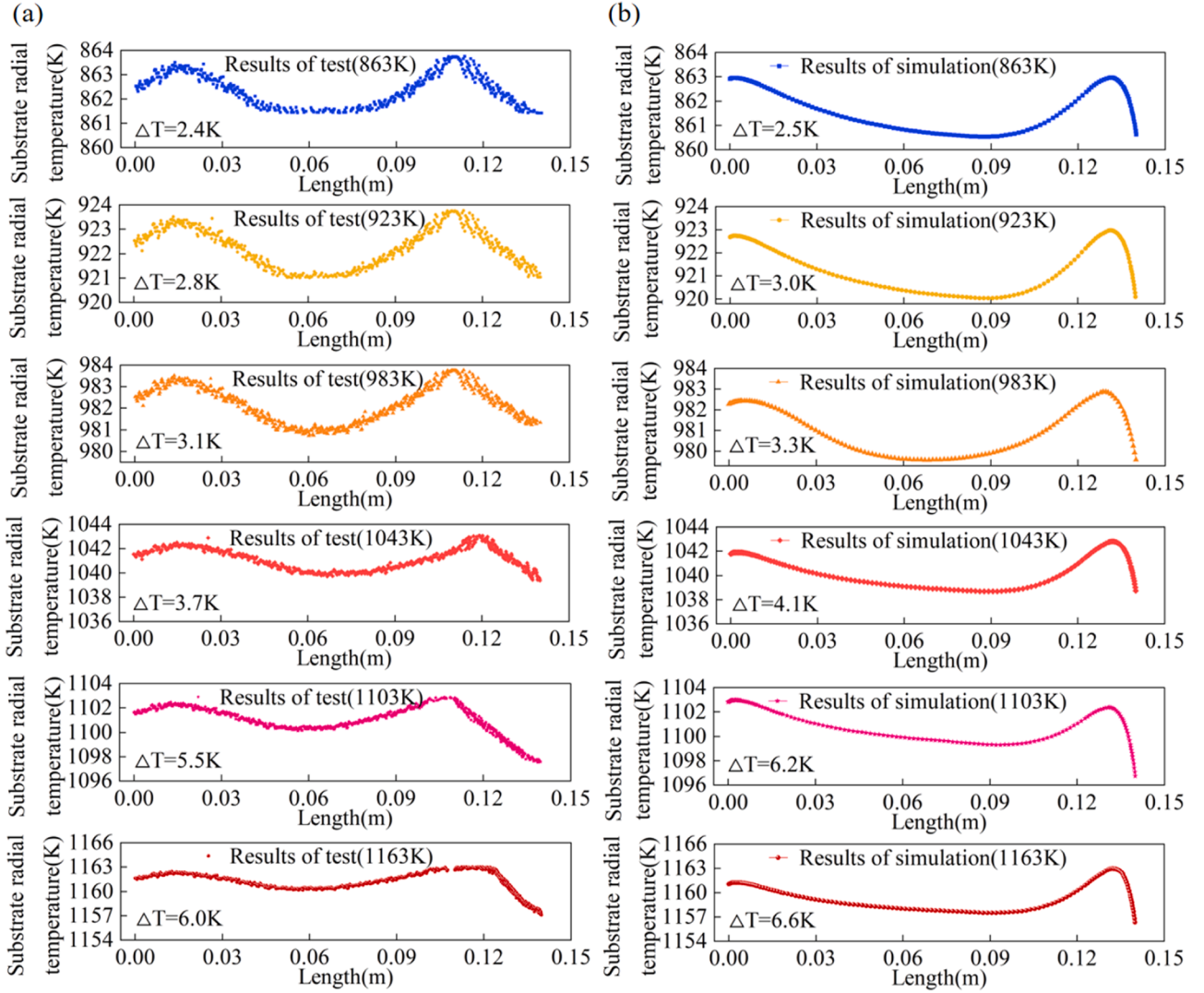


Fig. 14. Comparison of radial temperature distributions between experimental and simulation under different heating conditions. (a) Experimentally measured substrate radial temperature profiles at six target temperatures: 863 K, 923 K, 983 K, 1043 K, 1103 K, and 1163 K. (b) Corresponding simulated radial temperature profiles for the same heating conditions.

ensures the reliability of simulation-guided heater design and optimization, establishing a robust foundation for large-scale MBE implementation in high-uniformity heating systems.

4. Conclusions

This study presents a comprehensive thermal optimization framework for 4×6 -inch MBE systems, integrating design, simulation, and experimental validation of a novel dual-temperature zone heater. The proposed heater addresses the long-standing challenge of substrate temperature non-uniformity by combining dual-temperature zone coordinated control with multi-layer insulating and reflecting plates. Using a coupled multi-physics simulation validated by experiments, the heater achieves temperature uniformity within ± 3 K at a target temperature of 983 K and reaches a maximum temperature of 1275 K. The close agreement between simulation and experiment, with a maximum deviation of 2.4 K (relative error: 0.2 %), confirms the predictive accuracy of the multi-physics model.

A parametric study analyzed and quantified the heater's thermal behavior. Orthogonal experiments combined with nonlinear regression modeling revealed the influence of key parameters on temperature uniformity. The resulting predictive models demonstrated strong

statistical significance ($R^2 > 0.95$) and low predictive error ($< 5\%$), allowing for the accurate determination of optimal configurations. The heater adopts an independently controlled dual-temperature zone, substantially improving thermal field uniformity. Specifically, $L_A = 174$ mm and $L_B = 20$ mm significantly improve radial temperature distribution, achieving a peak uniform area ratio of 90.63 % and a 93.8 % enhancement in uniformity compared with a single-temperature zone.

Beyond the specific case analyzed, the proposed modeling and optimization framework provides a generalized theoretical basis for designing highly uniform thermal fields in MBE and other high-temperature processes. This study offers a referable methodology for multi-parameter co-design in complex thermal systems by linking structural parameters to thermal uniformity through simulation-validated predictive models.

CRediT authorship contribution statement

Yongshuan Wu: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Wenli Lv:** Supervision, Resources. **Fengwu Chen:** Writing – review & editing, Data curation. **Changping Chen:** Writing – review & editing. **Xin Gong:** Supervision, Project administration. **Junhui Li:** Supervision, Funding acquisition.

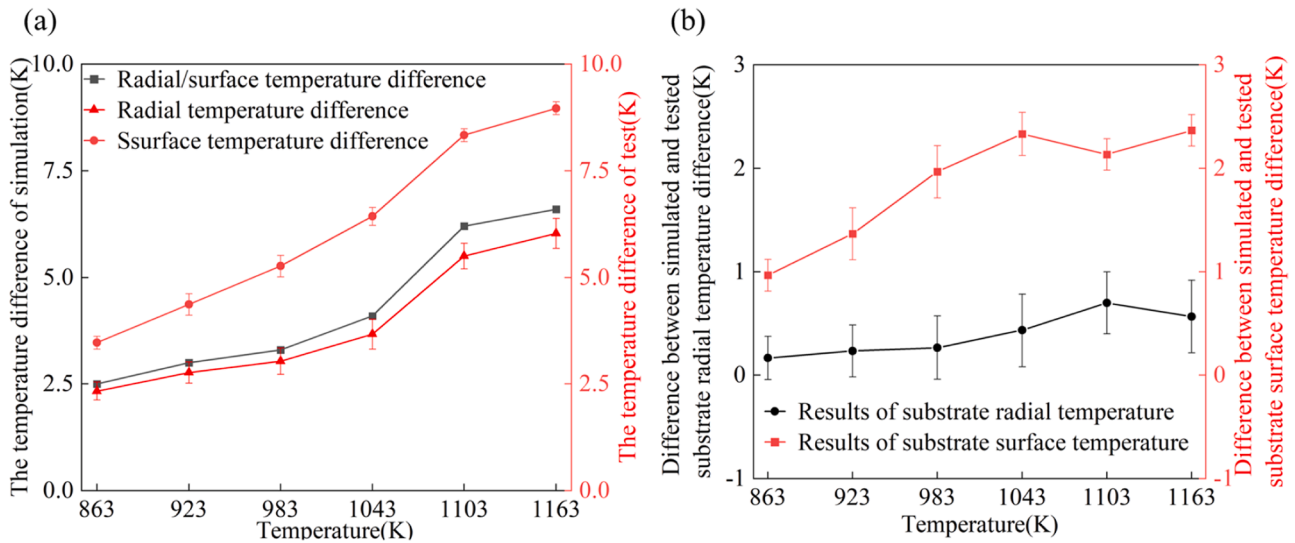


Fig. 15. Comparison between the tested and simulated values of substrate temperature difference under different target temperatures. (a) Radial/surface temperature difference of substrate. The black line represents the substrate's simulated radial/surface temperature difference, while the red line represents the radial/surface temperature difference of the substrate measured in the experiment. (b) The difference between the tested and simulated substrate is in the radial/surface temperature difference. The black line represents the difference in radial temperature; the red line represents the difference in surface temperature.

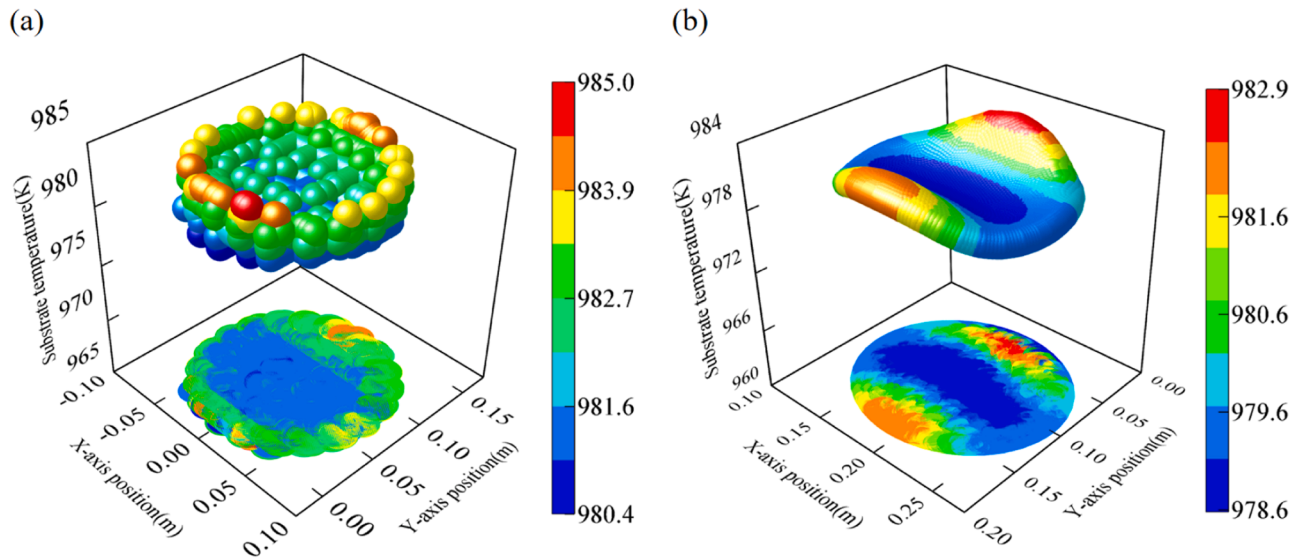


Fig. 16. 3D distribution of substrate surface temperature under dual-temperature zone heating. (a) Experimental measurement of the substrate surface temperature at 983 K. (b) Simulated surface temperature distribution at the same heating condition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

In this appendix, to facilitate understanding and improve readability, the key symbols, parameters, and abbreviations used throughout this manuscript are summarized in the following nomenclature table.

Nomenclature			
Symbols		q_{rad}	radiative heat flux, $W \cdot m^{-2}$
C_p	heat capacity, $J \cdot kg^{-1} \cdot K^{-1}$	Q_{joule}	joule heat, $W \cdot m^3$
d	width of heating plate, mm	R	range
e_b	radiosity	t	time, s
E	electric field strength, $V \cdot m^{-1}$	T	temperature, K
F_{12}	view factor from A_1 to A_2	T_{amb}	ambient temperature, K
F_{amb}	ambient view factor	V	electric potential, V
F_{ext}	external view factor	w	thickness of heating plate, mm
G	incident flux, $W \cdot m^{-2}$	Greek Symbols	
G_m	mutual radiation between different boundaries, $W \cdot m^{-2}$	β	substrate temperature difference, K
h_1	spacing of insulating plates, mm	γ	Effective radiation area, m^2
h_2	spacing of reflecting plates, mm	δ	average surface temperature of the shell, K
H	heating spacing, mm	ϵ	surface emissivity
I_i/J_i	sum of all evaluation indices for the i th level of a given factor	ϵ_{amb}	ambient emissivity
J	current density, $A \cdot m^{-2}$	λ	thermal conductivity
l	spacing of heating plate, mm	ρ	solid density, $kg \cdot m^{-3}$
L	width of under reflecting plate, mm	σ	conductivity
L_A	width of inner temperature zones, mm	σ_{SB}	boltzmann's constant
L_B	width of outer temperature zones, mm	Abbreviations	
n	refractive index	2DEG	two-dimensional electron gas
N_1	number of insulating plates	5G	5th generation mobile communication technology
N_2	number of reflecting plates	CVD	chemical vapor deposition
P	P value	MBE	molecular beam epitaxial
P_s	aggregate sum of radiation sources located at a point	PBN	pyrolytic boron nitride
q_{con}	conducted heat flux, $W \cdot m^{-2}$	PG	pyrolytic graphite
$q_{0,s}$	directional radiation sources	IoT	internet of things
		VCSEL	vertical-cavity surface-emitting lasers

Appendix B

This appendix presents supplementary figures. Provided supplementary information on the heater structural parameters, optimization process, simulation setup, and experimental measurement methods. These figures provide essential context for understanding the simulation and experimental procedures described in the manuscript [Figs. B1–B5](#).

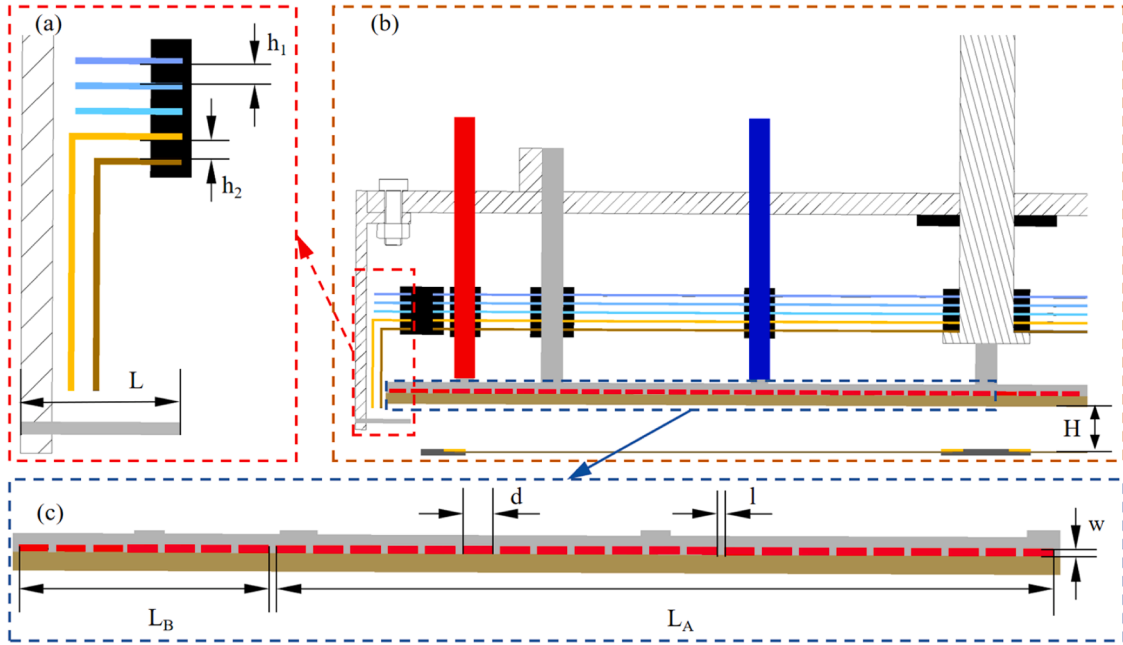


Fig. B1. Schematic diagram of structural parameters. (a) Definition of the spacing of insulating plate h_1 , the spacing of reflecting plate h_2 , and the width under reflecting plate L . (b) The heating spacing H represents the gap between the substrate and the heating plate. (c) Illustration of the heating plate dimensions and temperature zone configurations.

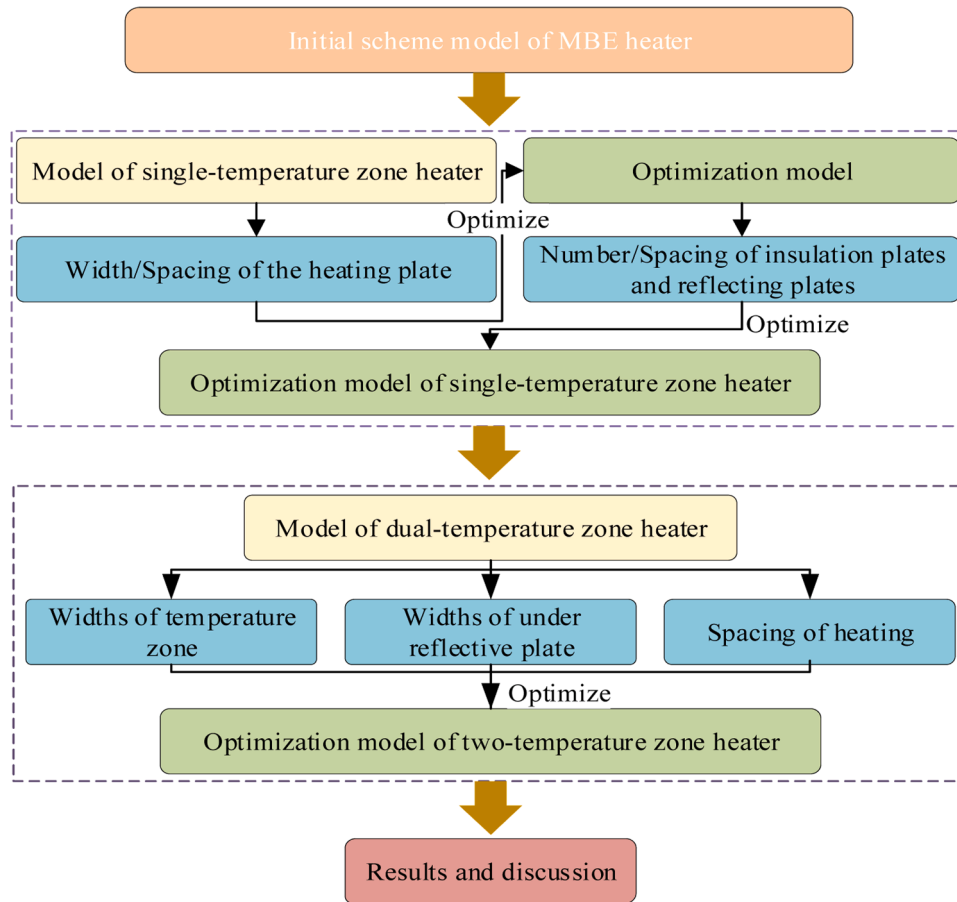


Fig. B2. The optimization process of the heater structure.

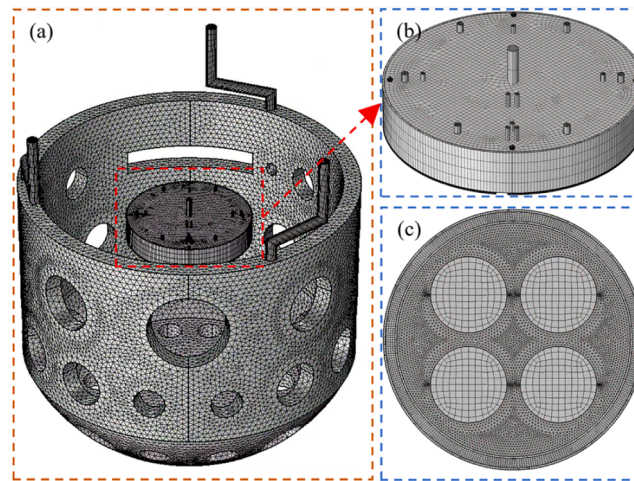


Fig. B3. Mesh model. (a) Mesh model of the MBE thermal management system; (b) Mesh model of the heater assembly; (c) Mesh model of the substrate. Tetrahedral meshes were employed for irregular geometries, while regular cylindrical, toroidal, and substrate structures were used with hexahedral meshes.

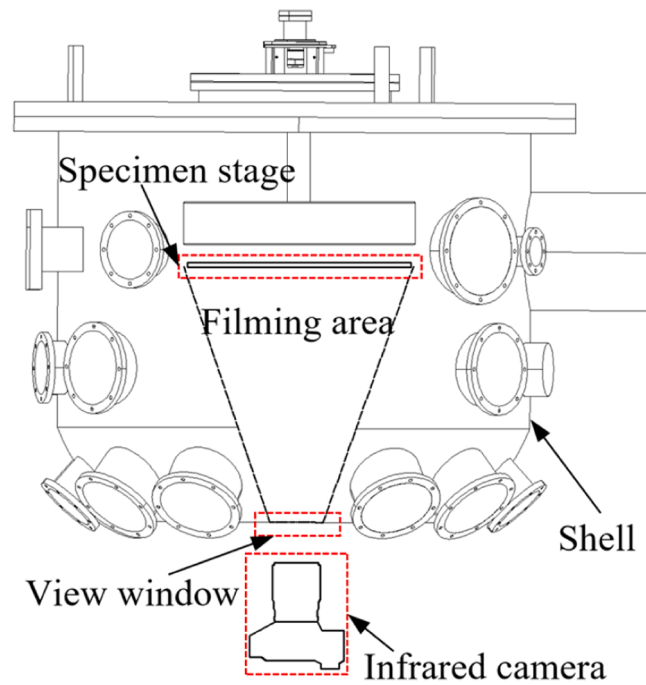


Fig. B4. Schematic diagram of temperature measurement. The infrared camera is installed at the bottom of the device, directly under the view window, to take real-time infrared photos of the heating process of the specimen stage.

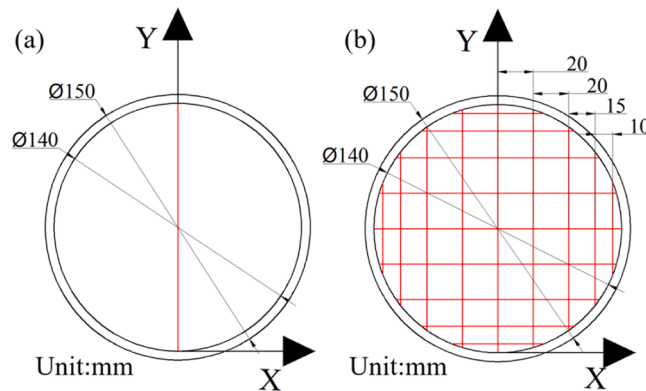


Fig. B5. Schematic diagram of sampling. (a) Radial substrate sampling; (b) Substrate surface temperature sampling. The red line in the figure shows the sampling position, and the black numbers indicate the substrate size and sampling spacing.

Appendix C

This appendix provides detailed tables containing quantitative results from the simulation and statistical analyses conducted in this study. These tables offer comprehensive numerical data supporting reproducibility and further analysis [Tables C1–C8](#).

Table C1
Key structural material parameters of heater.

Structure	Materials	Solid density $\rho(\text{kg}\cdot\text{m}^{-3})$	Thermal conductivity $\lambda(\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1})$	Heat capacity $C_p(\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1})$	Conductivity $\sigma(\Omega^{-1}\cdot\text{m}^{-1})$	Surface emissivity $\varepsilon(0-1)$
Heating plate	Pyrolytic graphite(PG)	1950	Fig. 2(b)	Fig. 2(b)	Fig. 2(a)	0.83
PBN board	Pyrolytic boron nitride(PBN)	1850	58	710	–	0.88
Platen,Backing ring	Mo (Rough Surface)	10,200	138	250	–	0.45
Insulating plate	Mo (Polished surface)	10,200	138	250	–	0.2
Reflecting plate	Ta (Polished surface)	16,650	57	140	–	0.1
Hull,Upper cover	Ta (Rough Surface)	16,650	57	140	–	0.5
Substrate	GaAs	5500	46	330	–	0.8

Table C2
Key parameters of the heater.

Type	Parameter	Unit	Value
Parameter analysis	width of heating plate (d)	mm	3, 5, 7, 9, 11
	spacing of heating plate (l)	mm	1, 2, 3, 4, 5
	thickness of heating plate (w)	mm	2
	number of insulating plates (N_1)	—	1, 2, 3
	spacing of insulating plate (h_1)	mm	1, 3, 5
	number of reflecting plates (N_2)	—	1, 2, 3
	spacing of reflecting plate (h_2)	mm	1, 3, 5
	width of under reflecting plate (L)	mm	16, 18, 20, 22, 24, 26, 28
	width of inner temperature zone (L_A)	mm	185, 174, 163, 152, 141
	width of outer temperature zone (L_B)	mm	9, 20, 31, 42, 53
	heating spacing (H)	mm	10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20

Table C3
Level table of factor about heating plate.

Levels	Factors		
	Width of heating plate d / (mm)	Spacing of heating plate l / (mm)	Error E
1	3	1	1
2	5	2	2
3	7	3	3
4	9	4	4
5	11	5	5

Table C4
Range analysis results of heating plate simulation.

Evaluation index		d	l	E	Evaluation index		d	l	E
β	I_1	586.2	597.5	582	γ	J_1	0.3495	0.5586	0.4625
	I_2	585.7	588.4	587		J_2	0.4253	0.4928	0.4522
	I_3	584.2	584.5	592		J_3	0.4707	0.4432	0.4577
	I_4	581	574.5	597		J_4	0.5009	0.4032	0.4472
	I_5	576.6	568.8	602		J_5	0.5223	0.3708	0.4491
R_i		9.6	28.7	20	R_j		0.1728	0.1878	0.0154

Table C5
F-test results of heating plate simulation.

Evaluation index		Source of variance	F value	F critical value	P value	Significance
β	d		14.047	$F_{0.01}(4,16) = 4.77$	< 0.0001	***
	l		113.809	$F_{0.05}(4,16) = 3.01$	< 0.0001	***
γ	d		92.531	$F_{0.1}(4,16) = 2.33$	< 0.0001	***
	l		108.097		< 0.0001	***

Table C6
Level table of factor about insulating plate and reflecting plate.

Levels	Factors				
	Number of insulating plates N_1	Spacing of insulating plate h_1 / (mm)	Number of reflecting plates N_2	Spacing of reflecting plate h_2 / (mm)	Error E
1	1	1	1	1	1
2	2	3	2	3	2
3	3	5	3	5	3

Table C7

F-test results of insulating plate and reflecting plate simulation.

Evaluation index	Source of variance	F value	F critical value	P value	Significance
β	N_1	1.68	$F_{0.01}(2,9) = 8.02$	0.2399	—
	h_1	0.489	$F_{0.05}(2,9) = 4.26$	0.6286	—
	N_2	36.811	$F_{0.1}(2,9) = 3.01$	< 0.0001	***
	h_2	8.793		0.0076	***
δ	N_1	5.105		0.0330	**
	h_1	1.072		0.3823	—
	N_2	23.682		0.0003	***
	h_2	0.323		0.7320	—

Table C8

F-test results of regression model.

Evaluation index	Source of variance	F value	F critical value	P value	Significance
β_1	$\beta_1 = -0.034d^2 + 0.236d + 116.83$	$F(2,2) = 2587.05$	$F_{0.01}(2,2) = 99$	0.0004	***
β_2	$\beta_2 = -1.426l + 120.83$	$F(2,2) = 222.87$	$F_{0.05}(2,2) = 19$	0.0045	***
β	$\beta = -0.034d^2 + 0.237d - 1.426l + 121.151$	$F(3,22) = 498,825.68$	$F_{0.01}(2,4) = 18$	< 0.0001	***
γ_1	$\gamma_1 = -0.0004d^2 + 0.01d + 0.043$	$F(2,2) = 393.37$	$F_{0.05}(2,4) = 6.94$	0.0025	***
γ_2	$\gamma_2 = 0.001l^2 + 0.016l + 0.126$	$F(2,2) = 2912.57$	$F_{0.01}(2,8) = 8.65$	0.0003	***
γ	$\gamma = -0.00035d^2 + 0.009d + 0.001l^2 - 0.016l + 0.083$	$F(3,22) = 8875$	$F_{0.05}(2,8) = 4.46$	< 0.0001	***
β	$\beta = 0.0124L^2 - 4.777L + 52.46$	$F(2,4) = 151.59$	$F_{0.01}(3,22) = 4.82$	0.0002	***
β	$\beta = 0.471H^2 - 14.72H + 123.48$	$F(2,8) = 114.62$	$F_{0.05}(3,22) = 3.05$	< 0.0001	***

Appendix D

This appendix presents the regression model building. Statistical tests and residual analyses confirm the accuracy, reliability, and absence of over-fitting, providing a solid basis for heater design optimization.

Separate univariate regression models were developed to analyze the relationships between substrate temperature difference, effective radiating area, heating plate width, and spacing. These univariate models were then integrated using multiple linear regression to develop the final multivariate nonlinear regression model [105].

$$\beta_1 = -0.034d^2 + 0.236d + 116.83, \quad (D1)$$

$$\beta_2 = -1.426l + 120.83, \quad (D2)$$

$$\gamma_1 = -0.0004d^2 + 0.01d + 0.043, \quad (D3)$$

$$\gamma_2 = 0.001l^2 - 0.016l + 0.126, \quad (D4)$$

$$\beta = \zeta_0 + \zeta_1\beta_1 + \zeta_2\beta_2 + \dots + \dots \sum_{i \geq 1}^{n \geq 1} \zeta_i\beta_n, \quad (D5)$$

where ζ_i is polynomial undetermined coefficients, β_n is optimal univariate regression models for each operational parameter and β , i is undetermined coefficients, n is the number of optimal univariate regression models.

Ultimately, the multivariate nonlinear regression models of β / γ are obtained as follows:

$$\beta = -0.034d^2 + 0.237d - 1.426l + 121.151, \quad (D6)$$

$$\gamma = -0.00035d^2 + 0.009d + 0.001l^2 - 0.016l + 0.083. \quad (D7)$$

As shown in Fig. 11(a), the substrate temperature difference (β) exhibits a non-monotonic trend as the width of under reflecting plate (L) increases, following a parabolic pattern where it first decreases and then increases. The substrate temperature difference reaches its minimum of 7.2 K at $L = 20$ mm. A quadratic function fit (Eq. (D8)) reveals that this trend can be accurately represented as:

$$\beta = 0.0124L^2 - 4.777L + 52.46. \quad (D8)$$

As shown in Fig. 11(b), the substrate temperature difference exhibits a parabolic trend as the heating spacing increases. The simulations achieved the minimum substrate temperature difference of 7.2 K when the heating spacing was 15 mm. A functional relationship between the substrate temperature difference (β) and heating spacing (H) was derived, with the specific equation provided in Eq. (D9).

$$\beta = 0.471H^2 - 14.72H + 123.48. \quad (D9)$$

An F-test was performed to assess the overall validity of the regression models, and P-values were used to evaluate the significance of individual variables. As shown in Table C8, all models passed the F-test ($P < 0.01$), indicating strong statistical relationships. These results confirm that the selected variables are highly correlated with the response and that the regression equations effectively capture variation trends within the experimental range.

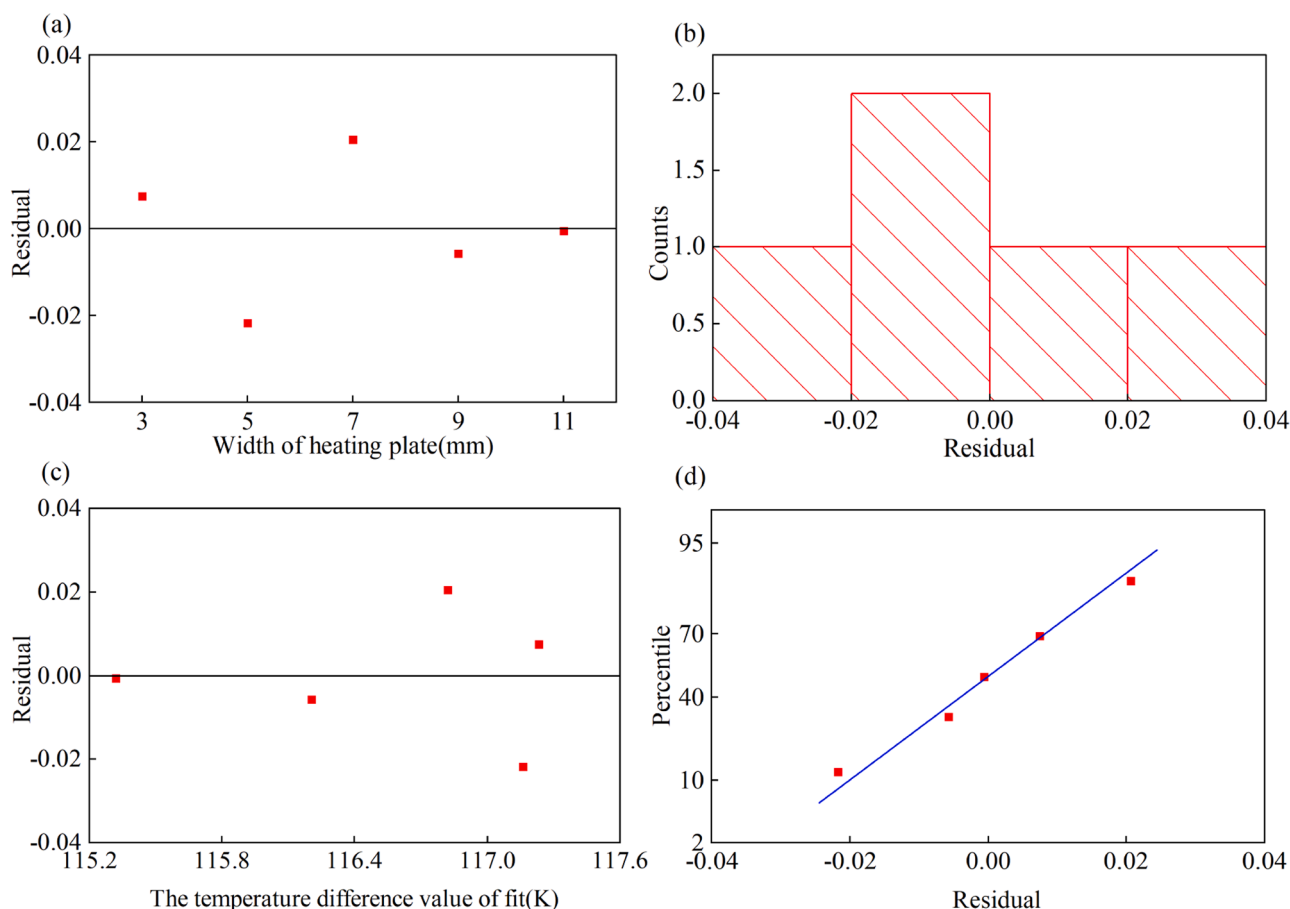


Fig. D1. Residual diagnostic plots for the fitted model of heating plate width (d) versus substrate temperature difference (β): (a) standardized residuals vs. heating plate width; (b) histogram of standardized residuals; (c) standardized residuals vs. fitted substrate temperature difference; and (d) normal probability plot (Q-Q plot).

Furthermore, A systematic diagnostic analysis of the fitted models was performed using standardized residual plots, histograms, and normal probability (Q-Q) plots. The residual distributions are shown in Fig. D1. The residuals exhibit a random distribution across variables and fitted values, with no evident patterns or systematic bias, indicating the absence of nonlinear trends or heteroscedasticity. The residual histogram approximates a normal distribution, and Q-Q plot points lie close to the reference line, confirming normality. These results demonstrate that the models fit well and satisfy the fundamental assumptions of regression analysis.

To further evaluate potential overfitting, residual plots were examined. As shown in Fig. D1, the residuals are randomly distributed with no discernible trends, and most standardized residuals fall within ± 2 , indicating minor model errors and approximate normality. Moreover, the fitted curves remain smooth without abrupt oscillations, suggesting appropriate model complexity and no evidence of overfitting.

Data availability

Data will be made available on request.

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